

AIT312 Recommendation System - Comprehensive Question Bank

MODULE 1: Introduction to Basic Concepts and Recent Developments

Part A Questions (3 Marks Each)

1. List any two purposes of recommendation system? (Jan 2024)
2. How is a collaborative recommendation system categorized? (Jan 2024)
3. Indicate the main idea of collaborative recommendation approaches. (June 2023, Dec 2024) ⚠
Repeated
4. Discuss the cases in which content-based recommendations will not perform as well as collaborative filtering. (June 2023)
5. How the data can be pre-processed for item based filtering? (May 2024)
6. Write a note on collaborative recommendation? (May 2024)
7. What inquiries will be tackled regarding the recent development in recommendation systems? (May 2024)
8. What is the central concept behind collaborative recommendation? (Dec 2024)
9. Describe the approaches for implicit and explicit ratings in collaboration system. (Dec 2024)
10. Explain what is TF-IDF? (Jan 2024)
11. With an example, explain working of Recommendation Algorithm? (From qp6)

Part B Questions (14/7 Marks)

14 Marks Questions:

1. Describe the two types of outputs generated with pure collaborative approaches that takes matrix of given user-item ratings as the only input? (Jan 2024)
2. Discuss content-based recommendation, in light of explicitly asking the user for his or her interests? (June 2023)
3. Write a note on cosine-similarity and what is the application of cosine-similarity in Recommendation system? (May 2024)
4. Explain similarity based retrieval and other text classification methods? (Jan 2024)
5. Provide a detailed discussion on content based recommendation system. (Dec 2024)

7 Marks Questions:

1. Explain about Rocchio's relevance feedback method (May 2024)

2. Discuss the implicit and explicit rating mechanism in collaborative recommendation approaches. (May 2024, June 2023) ⚠️ **Repeated**
3. Describe about user-based nearest neighbour recommendation system which deals with new items for which no ratings exist. (June 2023)
4. Elaborate on the user-based nearest neighbour recommendation system with an example. (Dec 2024)
5. Discuss the methods that tackle the issues of data sparsity and the cold start problem. (Dec 2024)

MODULE 2: Knowledge-based Recommendation

Part A Questions (3 Marks Each)

1. Define classical constraint satisfaction problem (CSP). (June 2023, Dec 2024, May 2024) ⚠️ **Repeated**
2. Differentiate between Dependent default and derived default? (June 2023, Jan 2024) ⚠️ **Repeated**
3. How can the performance of a vector space model be enhanced through various techniques and optimizations? (May 2024)
4. Explain knowledge based recommendation. (Dec 2024)
5. In what sense is Critiquing meant? (From qp6)
6. Explain TF-IDF. (From qp6)

Part B Questions (14/7 Marks)

14 Marks Questions:

1. Describe Critiquing algorithms. (June 2023, Jan 2024, Dec 2024) ⚠️ **Repeated**
2. How we can summarize the general interaction flow of knowledge based, conversational recommendation system? (May 2024)

7 Marks Questions:

1. How can we deal un-satisfiable requirements and empty results sets? (Jan 2024)
2. Explain QuickXPlain algorithm that calculates one conflict set at a time for a given set of constraints. (June 2023, Jan 2024) ⚠️ **Repeated**
3. Explain about the ranking of items/utility-based recommendation. (June 2023)
4. How defaults are utilized in constraint based recommendation? (Dec 2024)

MODULE 3: Hybrid Recommendation Approaches

Part A Questions (3 Marks Each)

1. What are the limitations of hybridization strategies? (June 2023, Jan 2024, May 2024, Dec 2024) ⚠️ **Repeated**
2. Explain about feature combination hybrids (June 2023, Jan 2024, May 2024) ⚠️ **Repeated**

3. Can we treat recommendation problem as utility function? (May 2024, Jan 2024) ⚠️ **Repeated**
4. What are the two dimensions of hybrid recommendation system? (Dec 2024)
5. What is meant by monolithic hybridization design? (From qp6)
6. Differentiate between parallelized hybridization design and pipelined hybridization design with suitable diagram. (From qp6)

Part B Questions (14/7 Marks)

14 Marks Questions:

1. Explain monolithic hybridization designs in detail. (Dec 2024)

7 Marks Questions:

1. Explain about feature combination hybrids. (Jan 2024, May 2024, June 2023) ⚠️ **Repeated**
2. Describe feature augmentation hybrid. (Jan 2024, May 2024, June 2023) ⚠️ **Repeated**
3. Explain about different parallelized hybridization strategies. (Jan 2024, June 2023, Dec 2024) ⚠️ **Repeated**
4. Differentiate between cascade hybrids and meta-level hybrids? (Jan 2024, June 2023) ⚠️ **Repeated**
5. Explain recommendation paradigm? (June 2023)
6. Explain how does parallelized hybridization work? (Dec 2024)
7. Explain about different pipelined hybridization design. (Dec 2024)

MODULE 4: Evaluating Recommender Systems

Part A Questions (3 Marks Each)

1. How can we distinguish between confidence and trust? (May 2024, Jan 2024) ⚠️ **Repeated**
2. What is the impact of Long Tail? (May 2024)
3. Explain evaluation paradigms? (June 2023, Jan 2024) ⚠️ **Repeated**
4. Differentiate RMSE versus MAE? (June 2023)
5. Explain A/B testing. (Dec 2024)
6. What is Impact of long tail? (From qp6)

Part B Questions (14/7 Marks)

14 Marks Questions:

1. Discuss about the general goals of evaluation design? (June 2023, May 2024, Jan 2024) ⚠️ **Repeated**
2. Explain the Accuracy metrics in offline evaluation? (May 2024)

7 Marks Questions:

1. Discuss about the design issues in offline recommender evaluation. Illustrate with a case study. (June 2023, Jan 2024, Dec 2024) ⚠️ **Repeated**
2. Explain about accuracy metrics in offline evaluation. (June 2023, Jan 2024, Dec 2024) ⚠️ **Repeated**
3. Explain about offline and online evaluations in recommender systems. (Jan 2024, May 2024) ⚠️ **Repeated**
4. Discuss about the various evaluation paradigms in recommendation. (Dec 2024)
5. Describe the general goals of evaluation design. (Dec 2024, Jan 2024) ⚠️ **Repeated**

MODULE 5: Attack-Resistant Recommender Systems

Part A Questions (3 Marks Each)

1. What is product push attack and nuke attack? (June 2023, Jan 2024) ⚠️ **Repeated**
2. Specify the effect of base recommendation algorithm. (Jan 2024)
3. Illustrate different methods used to detect attacks on existing recommendation system? (June 2023)
4. What are the different classifications of attacks? (May 2024)
5. How Fake profiles can be detected on Recommender systems? (May 2024)
6. What are the different methods in group attack profile detection? (Dec 2024)
7. What is probe attack? (Dec 2024)

Part B Questions (14/7 Marks)

14 Marks Questions:

1. Discuss about different attacks on recommender system? (June 2023, May 2024, Jan 2024, Dec 2024) ⚠️ **Repeated**
2. How do you quantify attack impact on recommender system? (June 2023, May 2024, Jan 2024) ⚠️ **Repeated**

7 Marks Questions:

1. Discuss about different methods available to detect attacks on recommender system (Jan 2024, Dec 2024) ⚠️ **Repeated**
2. Explain how to design robust recommendation algorithms. (Jan 2024, Dec 2024) ⚠️ **Repeated**
3. How to detect attacks on recommender system? (Dec 2024)
4. What are the different strategies to design a robust recommendation system? (Dec 2024)

Summary of Most Frequently Repeated Questions

Part A (3 Marks Questions):

1. Indicate the main idea of collaborative recommendation approaches.

2. Define classical constraint satisfaction problem (CSP).
3. Differentiate between Dependent default and derived default?
4. What are the limitations of hybridization strategies? ⚠️ **Most Repeated**
5. Explain about feature combination hybrids ⚠️ **Most Repeated**
6. Can we treat recommendation problem as utility function?
7. How can we distinguish between confidence and trust?
8. Explain evaluation paradigms?
9. What is product push attack and nuke attack?

Part B (14/7 Marks Questions):

1. Discuss the implicit and explicit rating mechanism in collaborative recommendation approaches.
2. Describe Critiquing algorithms. ⚠️ **Most Repeated**
3. Explain QuickXPlain algorithm that calculates one conflict set at a time for a given set of constraints.
4. Explain about feature combination hybrids. ⚠️ **Most Repeated**
5. Describe feature augmentation hybrid. ⚠️ **Most Repeated**
6. Explain about different parallelized hybridization strategies. ⚠️ **Most Repeated**
7. Differentiate between cascade hybrids and meta-level hybrids?
8. Discuss about the general goals of evaluation design? ⚠️ **Most Repeated**
9. Discuss about the design issues in offline recommender evaluation. Illustrate with a case study. ⚠️ **Most Repeated**
10. Explain about accuracy metrics in offline evaluation. ⚠️ **Most Repeated**
11. Explain about offline and online evaluations in recommender systems.
12. Discuss about different attacks on recommender system? ⚠️ **Most Repeated**
13. How do you quantify attack impact on recommender system? ⚠️ **Most Repeated**
14. Discuss about different methods available to detect attacks on recommender system
15. Explain how to design robust recommendation algorithms.

Module-wise Exam Weightage Analysis

Module	Part A Questions	Part B Questions	Total Questions	Percentage
Module 1	11	10	21	22%
Module 2	6	6	12	13%
Module 3	6	8	14	15%
Module 4	6	7	13	14%
Module 5	7	8	15	16%

Conclusion and Exam Preparation Strategy

Based on the analysis of the question papers from multiple years, the following topics deserve special attention:

1. **Module 1:** Focus on collaborative filtering concepts, user-based vs. item-based approaches, and rating mechanisms
2. **Module 2:** Pay special attention to Critiquing algorithms and the QuickXPlain algorithm
3. **Module 3:** Hybrid recommendation approaches, especially feature combination/augmentation and parallelized strategies
4. **Module 4:** Evaluation methods with focus on offline evaluation design and accuracy metrics
5. **Module 5:** Understanding different types of attacks and their impact on recommender systems

The most repeated questions (marked with ⚠️) should be prioritized in your preparation, as they have consistently appeared across multiple years and are likely to appear again.